chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC: ~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv

```
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~$ cd OS_Project
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project$ cd G4_Project1_xv6CustomizeSystemCalls
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls$ cd xv6-riscv
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv$ nano kernel/sysproc.c
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv$ nano kernel/syscall.h
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv$ nano kernel/syscall.c
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv$ nano user/user.h
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv$ nano user/usys.pl
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv$ nano user/testproc.c
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv$ nano Makefile
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv$ make clean
rm -f *.tex *.dvi *.idx *.aux *.log *.ind *.ilg \
*/*.o */*.d */*.asm */*.sym \
kernel/kernel fs.img \
mkfs/mkfs .gdbinit \
        user/usys.S \
user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie us
er/_logstress user/_forphan user/_dorphan user/_testhello user/_testproc user/_year_test
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv$ make
riscv64-linux-gnu-gcc -march=rv64gc -g -c -o kernel/entry.o kernel/entry.S
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o kernel/start.o kernel/start.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o kernel/console.o kernel/console.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o kernel/printf.o kernel/printf.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o kernel/uart.o kernel/uart.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o kernel/kalloc.o kernel/kalloc.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
```

chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC: ~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv

```
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o kernel/sysfile.o kernel/sysfile.c
riscv64-linux-gnu-gcc -march=rv64gc -g -c -o kernel/kernelvec.o kernel/kernelvec.S
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o kernel/plic.o kernel/plic.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o kernel/virtio_disk.o kernel/virtio_disk.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/kalloc.o kernel/spin
lock.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o
kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv$ make qemu
gcc -Wno-unknown-attributes -I. -o mkfs/mkfs mkfs/mkfs.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o user/ulib.o user/ulib.c
perl user/usys.pl > user/usys.S
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie -c -o user/usys.o user/usys.S
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o user/printf.o user/printf.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o user/umalloc.o user/umalloc.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o user/cat.o user/cat.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_cat user/cat.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_cat > user/cat.asm
riscv64-linux-gnu-objdump -t user/_cat | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/cat.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
```

chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC: ~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv

```
 -no-pie   -c -o user/dorphan.o user/dorphan.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_dorphan user/dorphan.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_dorphan > user/dorphan.asm
riscv64-linux-gnu-objdump -t user/_dorphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/dorphan.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o user/testhello.o user/testhello.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_testhello user/testhello.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_testhello > user/testhello.asm
riscv64-linux-gnu-objdump -t user/_testhello | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/testhello.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o user/testproc.o user/testproc.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_testproc user/testproc.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_testproc > user/testproc.asm
riscv64-linux-gnu-objdump -t user/_testproc | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/testproc.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o user/year_test.o user/year_test.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_year_test user/year_test.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_year_test > user/year_test.asm
riscv64-linux-gnu-objdump -t user/_year_test | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/year_test.sym
mkfs/mkfs fs.img README user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind
user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_testhello user/_testproc user/_year_test
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1011 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device
virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
 testproc
PID     STATE
        2
        2
        4
 QEMU: Terminated
chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC:~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv$
```

chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC: ~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv

GNU nano 7.2 kernel/sysproc.c

```
  return 0;
}

uint64
sys_kill(void)
{
  int pid;

  argint(0, &pid);
  return kkill(pid);
}

// return how many clock tick interrupts have occurred
// since start.
uint64
sys_uptime(void)
{
  uint xticks;

  acquire(&tickslock);
  xticks = ticks;
  release(&tickslock);
  return xticks;
}

uint64 sys_hello(void){
        printf("Hello from kernel");
        return 0;
}


uint64 sys_getprocinfo(void){
        struct proc *p;

        printf("PID\tSTATE\n");
        for(p = proc; p < &proc[NPROC]; p++){
                if(p->state != UNUSED) printf("%d\t%d\n", p->pid, p->state);
        }
        return 0;
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-Q Previous ^B Back
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was M-W Next ^F Forward

chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC: ~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv

GNU nano 7.2 kernel/syscall.h

```
#define SYS_fork    1
#define SYS_exit    2
#define SYS_wait    3
#define SYS_pipe    4
#define SYS_read    5
#define SYS_kill    6
#define SYS_exec    7
#define SYS_fstat   8
#define SYS_chdir   9
#define SYS_dup    10
#define SYS_getpid 11
#define SYS_sbrk   12
#define SYS_pause  13
#define SYS_uptime 14
#define SYS_open   15
#define SYS_write  16
#define SYS_mknod  17
#define SYS_unlink 18
#define SYS_link   19
#define SYS_mkdir  20
#define SYS_close  21

#define SYS_hello 23
#define SYS_getprocinfo 24
```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark	M-] To Bracket	M-Q Previous	^B Back
^X Exit	^R Read File	^\ Replace	^U Paste	^J Justify	^/ Go To Line	M-E Redo	M-6 Copy	^Q Where Was	M-W Next	^F Forward

chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC: ~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv

GNU nano 7.2 kernel/syscall.c

```
xtern uint64 sys_unlink(void);
xtern uint64 sys_link(void);
xtern uint64 sys_mkdir(void);
xtern uint64 sys_close(void);

xtern uint64 sys_hello(void);
xtern uint64 sys_getprocinfo(void);


/ An array mapping syscall numbers from syscall.h
/ to the function that handles the system call.
tatic uint64 (*syscalls[])(void) = {
[SYS_fork]    sys_fork,
[SYS_exit]    sys_exit,
[SYS_wait]    sys_wait,
[SYS_pipe]    sys_pipe,
[SYS_read]    sys_read,
[SYS_kill]    sys_kill,
[SYS_exec]    sys_exec,
[SYS_fstat]   sys_fstat,
[SYS_chdir]   sys_chdir,
[SYS_dup]     sys_dup,
[SYS_getpid]  sys_getpid,
[SYS_sbrk]    sys_sbrk,
[SYS_pause]   sys_pause,
[SYS_uptime]  sys_uptime,
[SYS_open]    sys_open,
[SYS_write]   sys_write,
[SYS_mknod]   sys_mknod,
[SYS_unlink]  sys_unlink,
[SYS_link]    sys_link,
[SYS_mkdir]   sys_mkdir,
[SYS_close]   sys_close,

[SYS_hello] sys_hello,
[SYS_getprocinfo] sys_getprocinfo,
};

void
syscall(void)
{
```

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GNU nano 7.2 user/user.h

```
#define SBRK_ERROR ((char *)-1)

struct stat;

// system calls
int fork(void);
int exit(int) __attribute__((noreturn));
int wait(int*);
int pipe(int*);
int write(int, const void*, int);
int read(int, void*, int);
int close(int);
int kill(int);
int exec(const char*, char**);
int open(const char*, int);
int mknod(const char*, short, short);
int unlink(const char*);
int fstat(int fd, struct stat*);
int link(const char*, const char*);
int mkdir(const char*);
int chdir(const char*);
int dup(int);
int getpid(void);
char* sys_sbrk(int,int);
int pause(int);
int uptime(void);

int hello(void);
int getprocinfo(void);

// ulib.c
int stat(const char*, struct stat*);
char* strcpy(char*, const char*);
void *memmove(void*, const void*, int);
char* strchr(const char*, char c);
int strcmp(const char*, const char*);
char* gets(char*, int max);
uint strlen(const char*);
void* memset(void*, int, uint);
int atoi(const char*);
int memcmp(const void *, const void *, uint);
```

[Read 52 lines]

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GNU nano 7.2 user/usys.pl

```
sub entry {
    my $prefix = "sys_";
    my $name = shift;
    if ($name eq "sbrk") {
        print ".global $prefix$name\n";
        print "$prefix$name:\n";
    } else {
        print ".global $name\n";
        print "$name:\n";
    }
    print " li a7, SYS_${name}\n";
    print " ecall\n";
    print " ret\n";
}

entry("fork");
entry("exit");
entry("wait");
entry("pipe");
entry("read");
entry("write");
entry("close");
entry("kill");
entry("exec");
entry("open");
entry("mknod");
entry("unlink");
entry("fstat");
entry("link");
entry("mkdir");
entry("chdir");
entry("dup");
entry("getpid");
entry("sbrk");
entry("pause");
entry("uptime");

entry("hello");
entry("getprocinfo");
```

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GNU nano 7.2 user/testproc.c

```
#include "kernel/types.h"
#include "user/user.h"

int main(){
        getprocinfo();
        exit(0);
}
```

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GNU nano 7.2 Makefile

```
# that disk image changes after first build are persistent until clean.  More
# details:
# http://www.gnu.org/software/make/manual/html_node/Chained-Rules.html
.PRECIOUS: %.o

UPROGS=\
	$U/_cat\
	$U/_echo\
	$U/_forktest\
	$U/_grep\
	$U/_init\
	$U/_kill\
	$U/_ln\
	$U/_ls\
	$U/_mkdir\
	$U/_rm\
	$U/_sh\
	$U/_stressfs\
	$U/_usertests\
	$U/_grind\
	$U/_wc\
	$U/_zombie\
	$U/_logstress\
	$U/_forphan\
	$U/_dorphan\
	$U/_testhello\
	$U/_testproc\
	$U/_year_test\

fs.img: mkfs/mkfs README $(UPROGS)
	mkfs/mkfs fs.img README $(UPROGS)

-include kernel/*.d user/*.d

clean:
	rm -f *.tex *.dvi *.idx *.aux *.log *.ind *.ilg \
	*/*.o */*.d */*.asm */*.sym \
	$K/kernel fs.img \
	mkfs/mkfs .gdbinit \
	$U/usys.S \
	$(UPROGS)
```

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